

PHARMACEUTICALS INORGANIC CHEMISTRY

UNIT - 1ST

◎ Syllabus: (Impurities) in Pharmaceutical Science

- History of Pharmacopoeia, Sources and types of Impurities, principle involved in the limit test for chloride, sulphate, iron, arsenic, lead and heavy metals, modified limit test for chloride and sulphate.

◎ Pharmaceutical Inorganic

Chemistry:— It is the branch of science in which we study about those inorganic compounds which involve in the preparation of drugs or medicines.

eg) NaCl, HCl (Acidifiers), Sodium Bicarbonate → Antacids
Copper sulphate (Ferments)

PHARMACOPOEIA

It can be defined as "a book of standards applicable to drugs and their common dosage forms and pharmaceutical aids published in a country under the authority of its own Government."

[OR]

The term pharmacopoeia comes from the greek word "pharmakon" means "drug" and "poeia" means "to make".

So,

It can be defined as "It is a standard official book which contains formulas (monographs) about to make a drug authorized by Government."

eg) I.P., B.P., U.S.P etc.
eg. Novo, Ingredients, Method of preparation etc. —→ It contains a complete details information of drugs.

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INDIAN PHARMACOPOEIA

It is an official document responsible for overall quality control and assurance of pharmaceutical products (pharmaceuticals) in India.

- It defines the identity, purity, safety and strength of drugs.
 - This book got approval under the second schedule of the Drug and Cosmetic Act 1940 and Rules 1945.
 - The Indian Pharmacopoeia is published by the Indian Pharmacopoeia Commission (IPC) on behalf of Ministry of Health and family welfare, Government of India.
 - The I.P. standards are legally acceptable.
- ### History and Development
- In 1833 firstly the first India company

recommended the publication of a pharmacopoeia and Bengal pharmacopoeia.

- from 1885 the B.P. was made official in India.
- After independence, in 1948 the Indian pharmacopoeia committee was developed to publish Indian pharmacopoeia. (IP).
- Now, the first edition of IP was published in 1955 i.e. IP 1955. by IPC
- It contains Monographs, which contains :-
 - List of drugs & related substances
 - Sources, tests, formulas
 - Uses, dosage and storage conditions etc..
- These information basically used as a reference standard for laboratory, industry and academic institutions.
- Now, editions of Indian Pharmacopoeia was published and several ~~some~~ addendum/supplement are performed :-

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INDIAN PHARMACOPOEIA EDITIONS

EDITIONS	YEAR	ADDENDUM OR SUPPLEMENT	No. of Volumes	MONOGRAPHS
• 1 st edition	1955	Supplement 1960	2	986
• 2 nd edition	1966	Supplement 1975	3	890
• 3 rd edition	1985	Addendum 1989 Addendum 1991	2	261
• 4 th edition	1996	Addendum 2000 Addendum 2002 Addendum 2005	3	1149
• 5 th edition	2007	Addendum 2008	3	271
• 6 th edition	2010	Addendum 2012	3	52
• 7 th edition	2014	Addendum 2015 Addendum 2016	4	577
• 8 th edition	2018	Addendum 2019	4	220

* 9th edition → 2022

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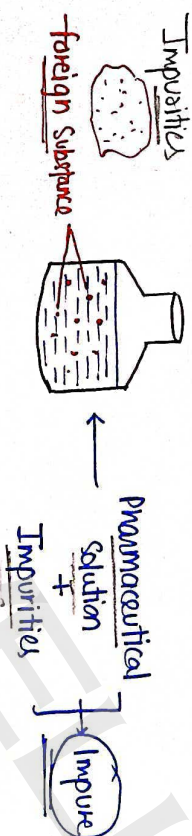
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• Impurities : SOURCES AND TYPES

- Impurities are unwanted foreign substances, which are present in pharmaceuticals and make it impure.
- Impurities may lower the quality of pharmaceuticals and also can cause toxicity.



eg) lead, calcium, arsenic, iron, chlorides etc--

• Types of Impurities :

They are mainly of three types -

- i) Organic Impurities
- ii) Inorganic Impurities
- iii) Residual solvent

i) Organic Impurities :

These are those impurities which produced during synthesis of drug substances, and during their storage.

- These impurities are basically "process and drug-related" i.e. it may also arise from raw-material, by-product, excipients, reagents etc--

- these impurities are not easily identified.

ii) Inorganic Impurities :

these are those impurities which basically produced during manufacturing process.

- It can be due to reagents, inorganic salts, heavy metal, catalyst etc--
- these impurities are easily identified by using limit test.

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iii) Residual Solvent:-

these are those impurities which produced during manufacturing process from solvents.

• Impurities may also categorised into two types on the basis of their effects:

i) Destructive Impurities:-

वैपरीत
These are those impurities which basically stop/prohibited the functions of substances.

• It may be removed by chemical method i.e. chemically and physical method i.e. distillation etc.

ii) Constructive Impurities:-

सहायक
These are those impurities which basically improves the functions of substances.

eg) Adding less than 2% carbon in pure iron to manufacture steel.

• Sources of Impurities:-

The impurities in drug products may be obtained from various sources such as during manufacturing, from raw material, excipients or accidentally.

• the most common sources are:-

- i) Raw Materials employed in Manufacture.
- ii) Reagents used
- iii) Processes or Method used
- iv) Environment related
- v) Defects
- vi) Manufacturing Hazards
- vii) Storage
- viii) Accidental / Adulteration.

i) Raw Materials → Sometimes, impurities occurred from raw material which contaminate the end product.
eg) traces of lead, copper, silver etc..

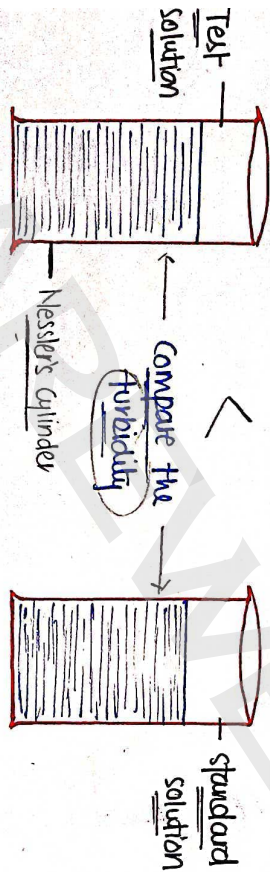
- (ii) Reagent used in manufacturing → impurities occurs from reagents if they are not washed properly and contaminate final product.
(eg) - Ammoniated mercury precipitate - impurity.
- (iii) Processes → During manufacturing, some impurities enter into the materials.
• It may be due to formulation/exipients.
• also arise due to residue solvent.
• due to chemical process. (eg) oxidation.
- (iv) Environment related → Sometimes, impurities occurred from atmosphere, these may cause contamination in final products. (eg) Gases - H_2S , SO_2 etc - temp, humidity.
- (v) Defects → Due to defects occurs during manufacturing such as improper mixing, pressure, pH etc.
- (vi) Manufacturing Hazards → Contamination occurs during manufacturing.
• Due to mixing of dirt, glass, metallic or plastic fragments.
• Due to gross error i.e. overall error.
- (eg) cross contamination, microbial contamination and packing errors.
- (vii) Storage → Sometimes, impurities are occurred during storage i.e. dust, insect bodies, animal etc. may pollute the products.
- (eg) chemical instability, physical change, temp.
- (viii) Accidental → Sometimes, some toxic substances are mixed accidentally which we not known.
(eg) expensive KBr is adulterated with cheaper $NaBr$.
- ③ Effects :-
A little amount of impurity always remain in a material, so if the amount of impurity is present in excess, then it cause toxicity :-
- cause toxic effects.
- alter the physical & chemical properties.
- make therapeutically inactive.
- reduce the shelf-life of a substances.
- may changes the color, odour, taste etc and making the substance unhygienic and unfit to use.

LIMIT TESTS

These are quantitative or semi-quantitative test designed to identify and control small quantity of impurities which are lightly present in substances.

[OR]

These are those tests which are basically used for determination of amount of impurities. (eg) limit test for chloride etc.



• It is based on the principle that in this, we compare the turbidity/opalescent of test solution with standard solution.

• If the turbidity of test solution is less than standard solution, then limit test passed.

If, Turbidity of test solution < Turbidity of standard solution — Passed

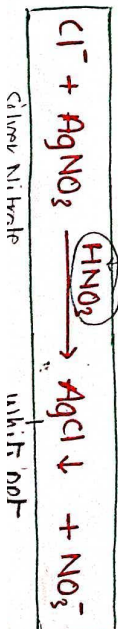
If, Turbidity of test solution > Turbidity of standard solution — Failed

LIMIT TEST FOR CHLORIDE

It is basically used for identifying the chloride ion impurities and measure their quantity.

• Principle:

It is based on the reaction, in which silver nitrate is react with soluble chloride which form silver chloride which is insoluble in dil. nitric acid. precipitates



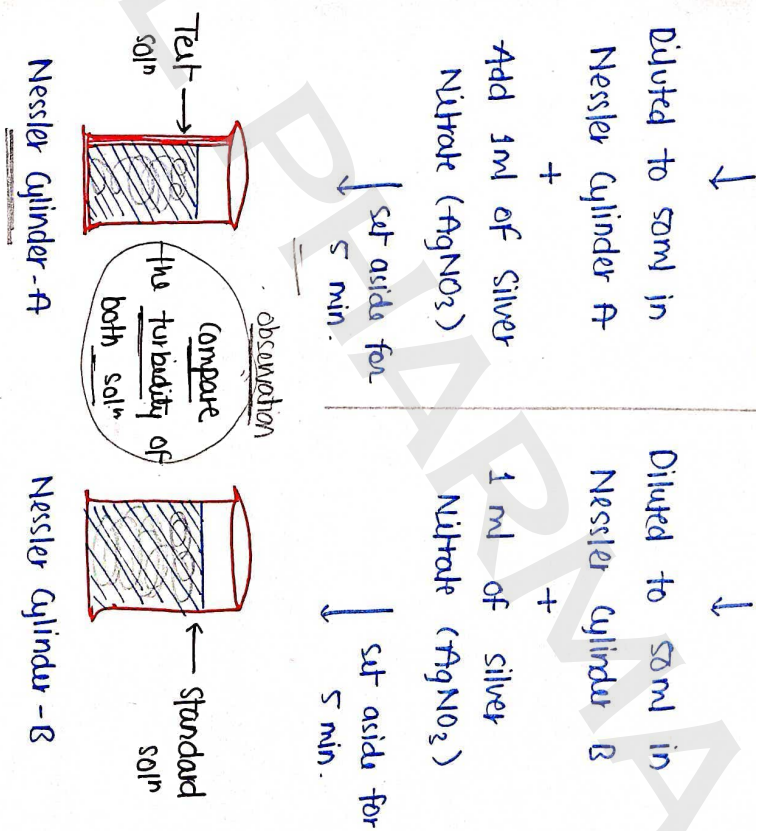
- Now, test solution appears turbidity due to formation of silver chloride (ppt)
- Degree of turbidity is depends on the amount of chloride, which then compared with standard solⁿ.

Procedure :

for this, we have to prepare test as well as standard solution.

- Apparatus : Nessler's cylinder, Measuring cylinder, Glass rod etc. - same for all limit-test
- Chemicals : Dil. Nitric acid, silver Nitrate, NaCl.

TEST SOLUTION	STANDARD SOLUTION
(Nessler Cylinder - A)	(Nessler Cylinder - B)
Specified substance (1g)	1ml of 0.0584% w/v sol ⁿ of sodium chloride
+ 10 ml of water	+ 1ml of Nitric acid
+ 1ml of Nitric acid (HNO ₃)	(HNO ₃)



Result :

- If the turbidity of test solution is less than the turbidity of standard solution, then sample passed the limit test and ready for use
- If the turbidity of test solution is more than the turbidity of standard solution, then sample failed the limit test and need to control

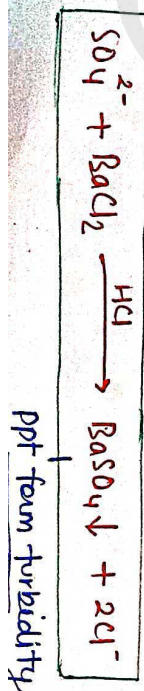
• Why dil. HNO_3 (Nitric acid) used →
 It is used to make solution acidic, so they dissolve other impurities except chloride, because they are insoluble in HNO_3 and make it turbid.

LIMIT TEST FOR SULPHATE

This test is basically used for controlling and identifying the sulphate impurity in Substance.

• Principle:

It is based on the reaction, in which barium chloride is react with soluble sulphate in the presence of dil. HCl solⁿ which forms barium sulphate precipitate which generate turbidity.



then this turbidity is compared with std. solⁿ.
 • In this, barium chloride (BaCl_2) is present in the form of barium sulphate reagents, which contains, barium chloride, sulphate-free alcohol, potassium sulphate.
 (increase the sensitivity of test. prevent super saturation)

• Procedure:

In this, we have to prepare test as well as standard solution.

TEST SOLUTION (Nessler cylinder - A)	STANDARD SOLUTION (Nessler cylinder - B)
1 gm of specified substance of sulphate + 2 ml of HCl ↓ Added in 45 ml water + 5 ml of <u>barium sulphate</u> reagent	1 ml of 0.1089% w/v sol ⁿ of K_2SO_4 + 2 ml of HCl ↓ Added in 45 ml dist. water + 5 ml of <u>barium sulphate</u> reagent

Set aside for 5 min ↓ check the turbidity	Set aside for 5 min ↓ check the turbidity
--	--

- Test solution - 1gm of sulphate is weighed and 2ml of HCl is added to 45ml of distilled water, then 5ml of Barsoy reagent is added to prepare the solution.

- Standard solution - 1 ml of 0.1889% w/v solution of K_2SO_4 is weighed and treated with 2ml of HCl, then dilute it upto 45ml of water. At last add 5ml of Barsoy reagent. (15ml) of 0.5M $NaCl$ is mixed with (55ml) of water, then, add (20ml) of sulphate-free alcohol and (5ml) of 0.0181% w/v solution of potassium sulphate, then these all diluted upto (100ml) water.
- Result: If the turbidity of test solution is less than the turbidity of standard solution, then limit test passed.

• MODIFIED LIMIT TESTS FOR CHLORIDE AND SULPHATES

These are those tests which are basically used to overcome those problems which occurs during limit tests.

FOR CHLORIDE

- This test is basically used for colored substances, because sometimes we used any colored substance for test solution.

- These colored substance can create problem in formation of turbidity.

So, in modified limit test, we convert these colored substances into colorless substances, so, we get clear turbidity during test.

@g $KMnO_4$ (potassium permanganate).

In this, $KMnO_4$ gives purple color which create disturbance in turbidity.

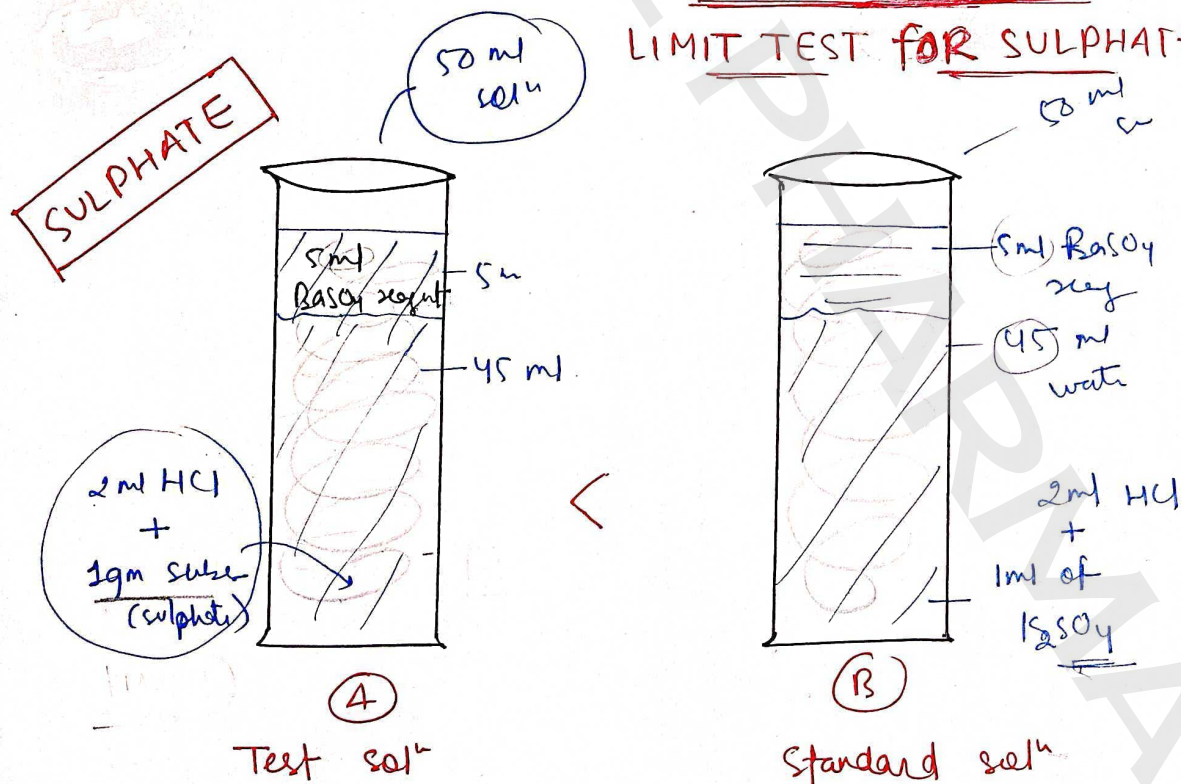
- for this, KMnO_4 is reacted with ethanol

$$2\text{KMnO}_4 + 2\text{C}_2\text{H}_5\text{OH} \xrightarrow[\text{redox rxn}]{\text{reducing agent (ethanol)}} 3\text{CH}_3\text{CHO} + 2\text{KOH} + 2\text{MnO}_2 + 2\text{H}_2\text{O}$$

purple color colorless magnesium oxide
- preparation of KMnO_4 : Dissolve 1.5 gm of KMnO_4 in 50 ml of dist. water, heat on water bath and add 6 ml of 95% ethanol. Cool and dilute upto 60 ml.
- then all procedure same as limit test for chloride.
- **FOR SULPHATE**

This test is also used for colored substance which create interference in turbidity i.e. KMnO_4 . (purple color) → reaction same as above (chloride). So, we convert it into colorless substance MnO_2 and follow further procedure of limit test for sulphate.

ROUGH DIAGRAM FOR LIMIT TEST FOR SULPHATE



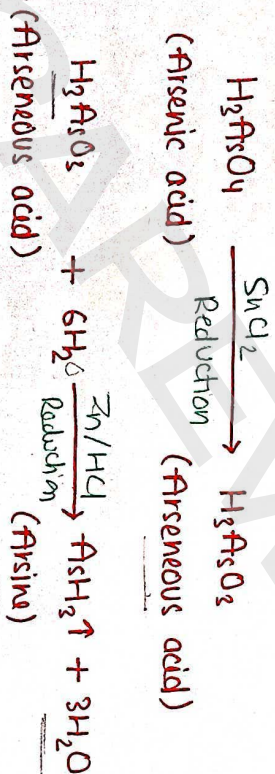
LIMIT TEST for ARSENIC

This test is used for identifying and controlling ^{arsenic} impurities in substances.

Arsenic is well known undesirable and harmful substance present in medicine substances.

• Principle:

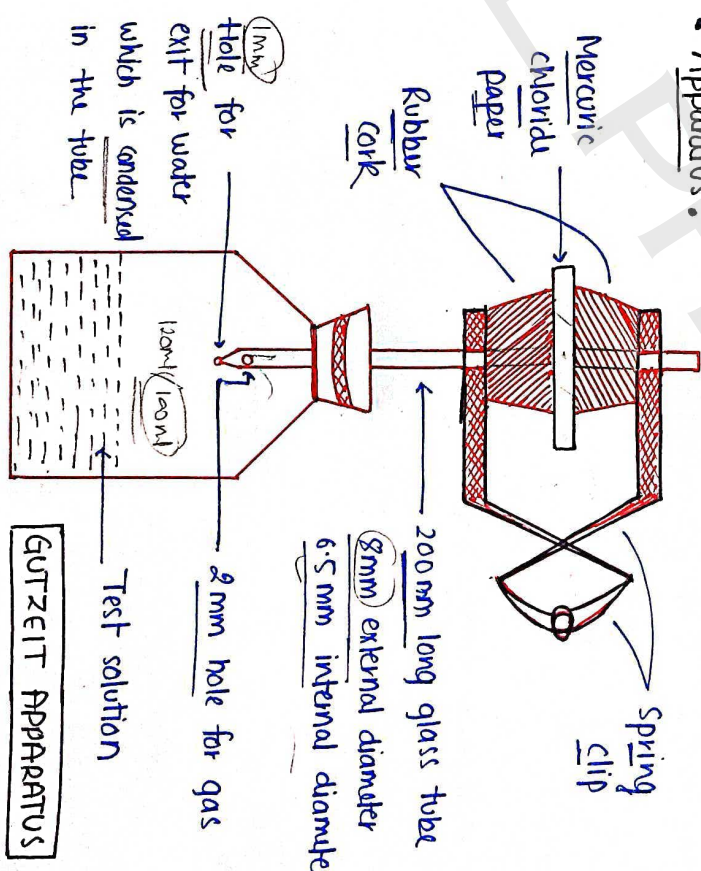
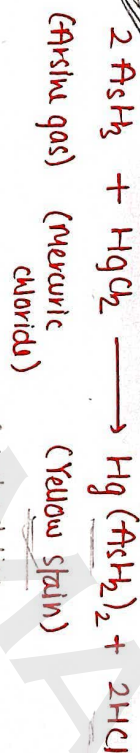
It is based on the reaction, in which firstly arsenic in arsenous acid state is converted into arsine gas by reduction with zinc and hydrochloride acid.



Then, this arsine gas passed through mercuric chloride paper and produce a yellow stain.

• Apparatus:

Now, these yellow stain is compared with (yellow stain) of standard solution. depends on intensity of these colors - arsenic quantity



- During test, the arsine gas passed through side hole and reached to mercuric chloride paper,

- Now, arsine gas produces yellow stain on mercuric chloride paper.
- The intensity of yellow stain is depends on the amount of arsenic impurities present in substances.
intensity of color $\propto$ arsenic present

• Procedure:

Prepared both test and standard solution and compared their intensity of colors.

TEST SOLUTION	STANDARD SOLUTION
- Test solution prepared and placed in generator bottle with HCl + Add (5ml) of 1M KI (potassium iodide) / (1gm) of KI + 5ml of stannous chloride SnCl_2 + 10 gm of zinc (AST) ^{arsenic free}	• 1ml of standard solution of (arsenic with HCl) + Add 5ml of 1M KI (potassium iodide) / 1gm of KI + 5ml of stannous chloride SnCl_2 + 10 gm of zinc (AST) ^{free from arsenic}

Keep the solution aside for 40 min.

↓

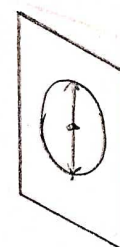
arsine gas react with HgCl_2 paper and produce yellow stain



Keep the solution aside for 40 min.

↓

arsine gas react with HgCl_2 paper and produce yellow stain



- compare these intensity/diameter of yellow stain.
- If intensity of yellow stain in test solution is less, than limit test passed. — **Result**

• Precautions:

- stain should be compared immediately as it fades away on standing.
- (arsenic chloride paper should be store in dark place)
- the tube should be washed with HCl (AST) than with water, then dry it.

LIMIT TEST for IRON

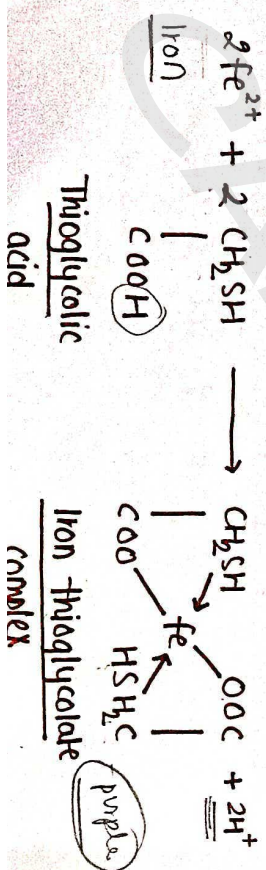
This limit test is used for identifying and controlling the iron impurities in solution.

• Principle :

It is based on the reaction in which iron is react with thioglycolic acid [mercapto acetic acid] in the presence of ammonium citrate buffer, which forms iron thioglycolate complex of a purple color.

Then the intensity of this purple color is compared with standard solution.

Amount of iron impurities $\propto$ intensity of purple color



• Procedure :

firstly, we have to prepared test as well as standard solution.

TEST SOLUTION	STANDARD SOLUTION
Sample is dissolved in water then volume is made up with 40ml.	2ml of standard solution of <u>iron</u> diluted with water upto 40 ml.
↓	↓
Add 2ml of 20% w/v citric acid [iron free]	Add 2ml of 20% w/v citric acid [iron free].
+	+
Add 2 drops of thioglycolic acid (mercapto acetic acid)	Add 2 drops of thioglycolic acid (mercapto acetic acid).
↓	↓
Add ammonia to make solution alkaline & make solution upto 50 ml	Add ammonia to make solution alkaline and make solution upto 50 ml.
keep aside for 5 min	keep aside for 5 min

Developed color is compared each other

• Observation / Result :

- If the intensity of purple color of test solution is less than standard solution, then sample is passed limit test.
- In this, the solution should be compared immediately within 5 minutes.

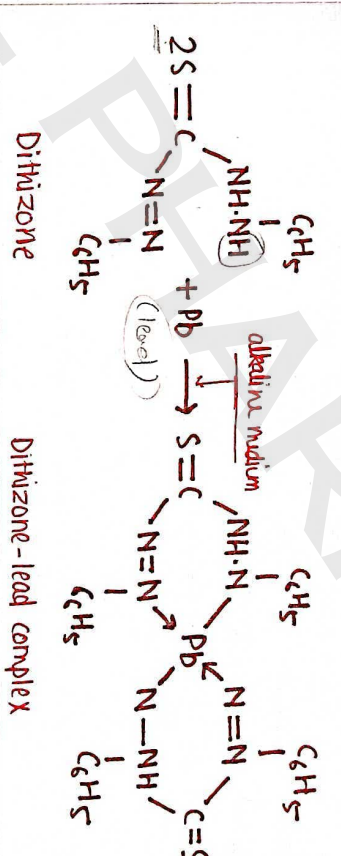
LIMIT TEST FOR LEAD

This limit test is used for identifying and controlling the lead impurities in solution.

• Principle :

It is based on the reaction, in which lead is react with diphenylthiocarbazone (dithizone), in alkaline solution to form lead dithizonate complex which is red in color.

and lead - dithizonate complex is violet in color, so the resulting color at the end is red.
At last we compare the color.



• Procedure :

In this, firstly we prepared test as well as standard solution, both solutions.

i) Test solution :

A known quantity of sample solution is transferred in a separating funnel.

Add 6ml of ammonium citrate ↓
40gm citric acid in 90ml water and ammoniacal solution

Add 2 ml of potassium cyanide and 2 ml of hydroxylamine hydrochloride

↓

Add 2 drops of phenol red and make solution alkaline by adding ammonia solution.

↓

Now, 5 ml dithizone solution is used to extract the above solution until it becomes green.

↓

Combine the dithizone extracts and transfer to a second separator.

↓

Shake the combined dithizone extracts for 30 seconds with 30 ml of 1% v/v

nitric acid solution and discard the chloroform layer.

↓

Now, in this acid solution add 5 ml of standard dithizone solution and also add 4 ml of ammonium cyanide

10 mg dithizone in 1000 ml of chloroform.

Now, shake the solution for 30 seconds and observe the color of chloroform layer.

↓

ii) Standard solution:

A standard solution of lead is transferred in a separating funnel.

↓

Now, follow the same procedure as test solution

• Observation/Result:

The color of chloroform layer developed in test solution should not be more intense than the color obtained in standard solution.

• Precaution:

All reagents/chemicals should be lead free. Glassware should be washed with dil. HNO_3 & water.

LIMIT TEST FOR HEAVY METALS

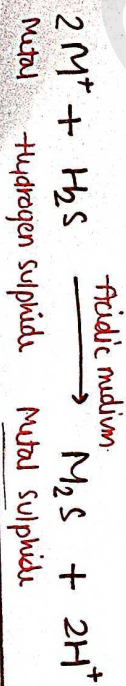
This limit test is basically used for determining the metallic impurities.

• Heavy Metals - these are those chemical elements which have high density and it is toxic or poisonous at low concentration.

• examples - Cadmium, lead (Pb), Mercury (Hg), Arsenic (As), Chromium (Cr), Zinc (Zn), Bismuth (Bi), Cobalt, Silver etc..

• Principle :

It is based on a reaction in which a solution of heavy metals (M) is reacted with Hydrogen Sulphide (H_2S) in an acidic medium, which forms reddish / brown color of metal sulphide.



Now, we compare the intensity of color of test solution with standard solution.

- In this test, we use lead as standard solution.
- It is expressed in 'lead parts per million' i.e. 20ppm for lead - test solution should not be more than this.

• Procedure :

Prepare test as well as standard solution

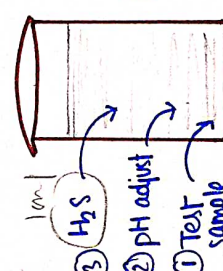
TEST SOLUTION	STANDARD SOLUTION
• Specific quantity of sample is dissolved in <u>25 ml</u> water in <u>50 ml</u> Nessler cylinder. ↓ • pH is adjusted between <u>3-4</u> by dil. acetic acid / dil. ammonia solution. ↓ • Volume upto <u>55 ml</u> water. ↓	• Take <u>2 ml</u> of standard <u>lead</u> solution in <u>25 ml</u> water in <u>50 ml</u> Nessler cylinder. ↓ • pH is adjusted between <u>3-4</u> by dil. acetic acid / dil. ammonia solution. ↓ • Volume upto <u>35 ml</u> water. ↓

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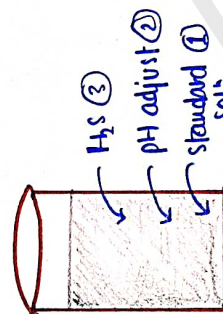
- It is also performed by other methods :
 - Method A : for those substances who gives clear colorless solution, H_2S used.
 - Method B : for those substances, which does not give clear colorless solution, H_2S used.
 - Method C : for those substances which gives clear colorless solution, HNO_3 used.
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- Add 10 ml of freshly prepared H_2S [Hydrogen Sulphide]
- Make volume upto 50ml by adding water.
- Keep aside for 5 min



- ③ H_2S
- ② pH adjust
- ① Test sample

- Add 10 ml of freshly prepared H_2S [Hydrogen Sulphide]
- Make volume upto 50ml by adding water.
- Keep aside for 5 min



- ③ H_2S
- ② pH adjust
- ① Standard sol

Compare the intensity of color of both solutions.

Observation / Result :

If the color produced in test solution is less than the color produced in standard, then limit test passed.

Color produced in Test solution	<	Color produced in Standard solution
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